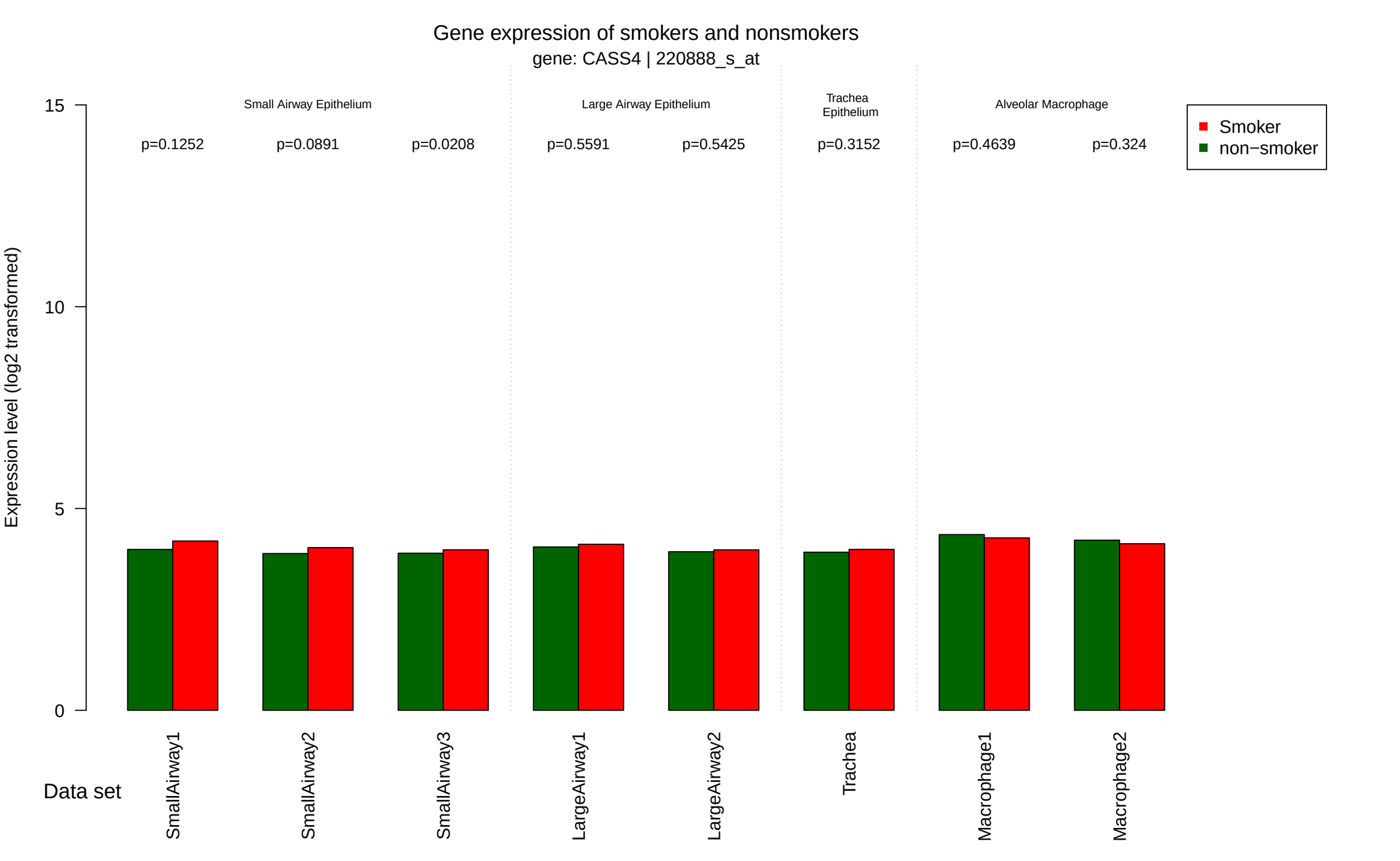
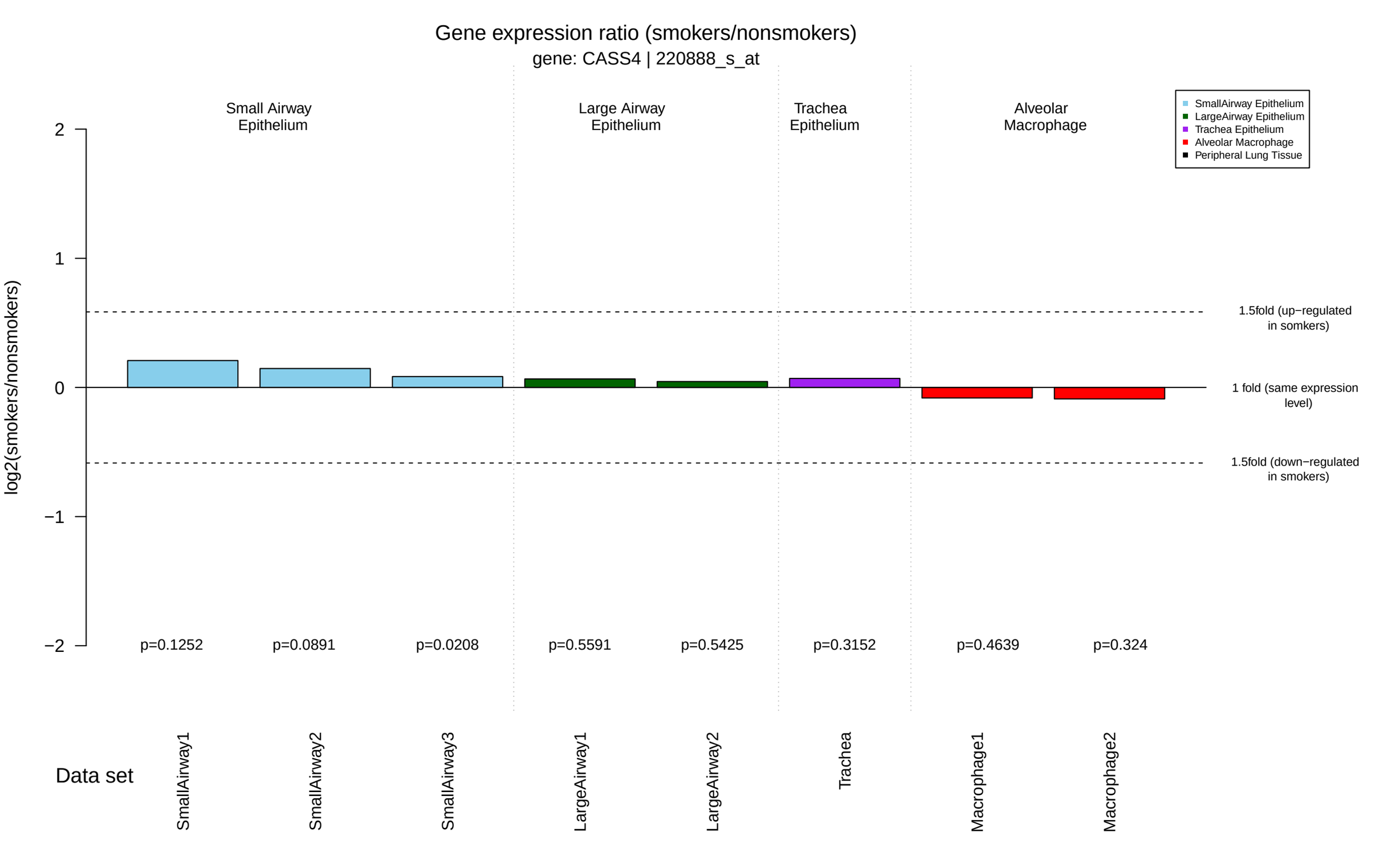
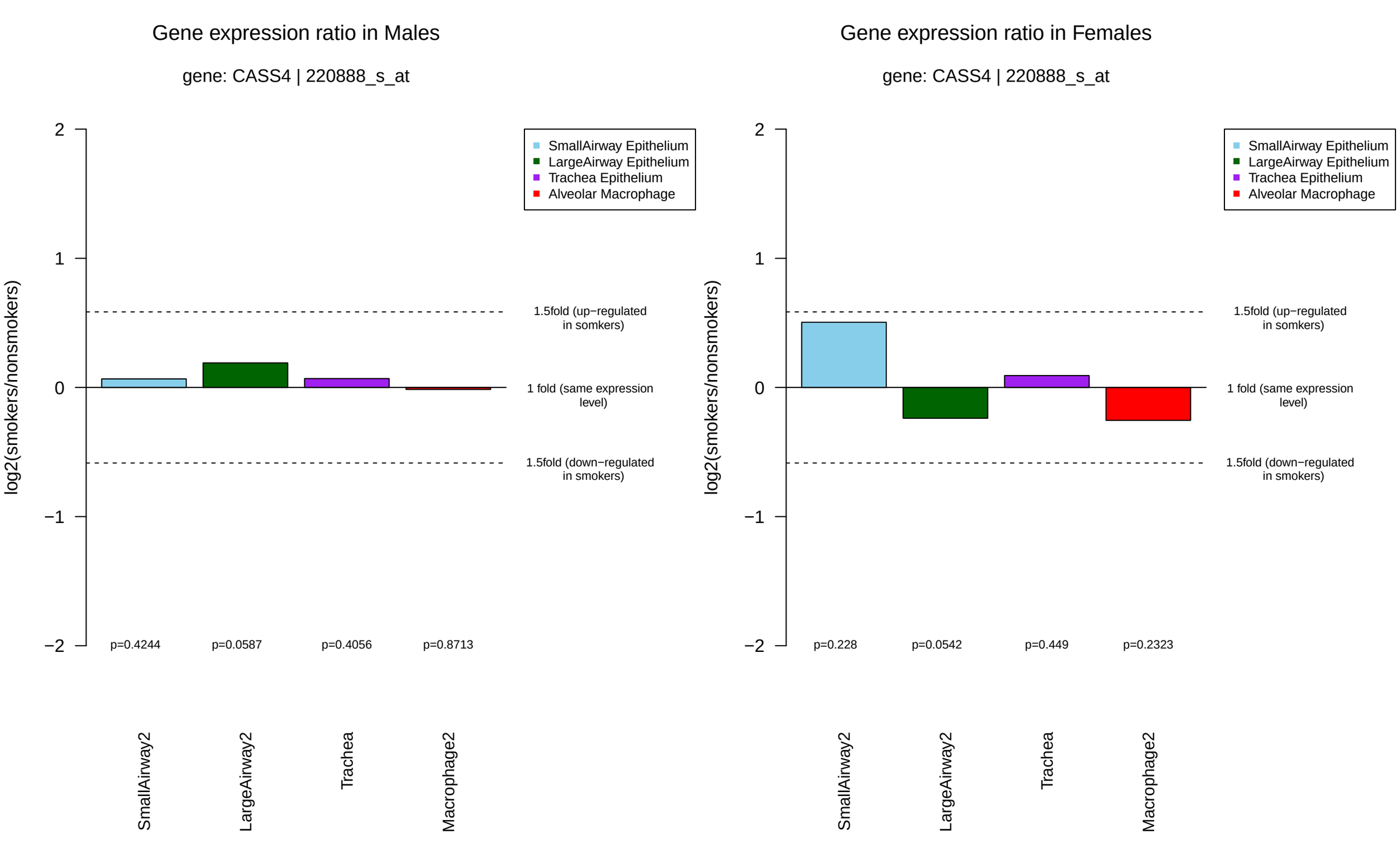
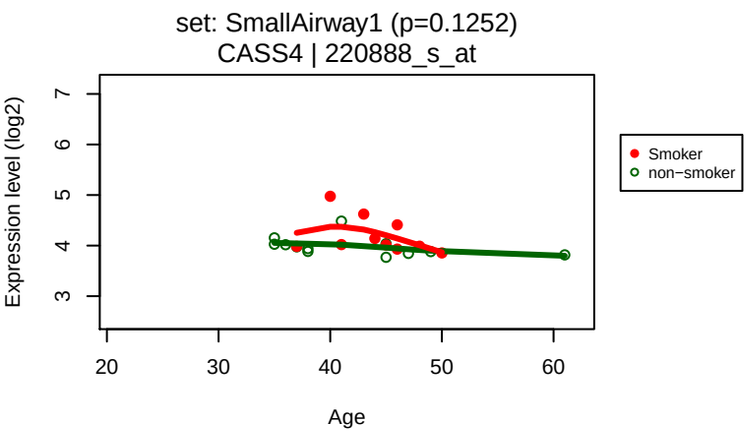
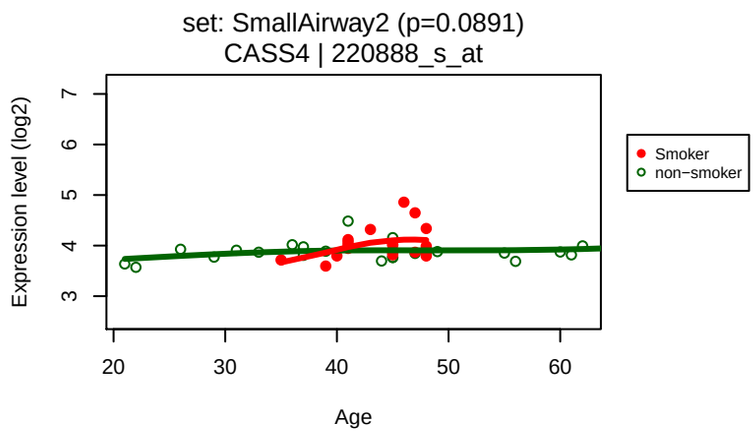
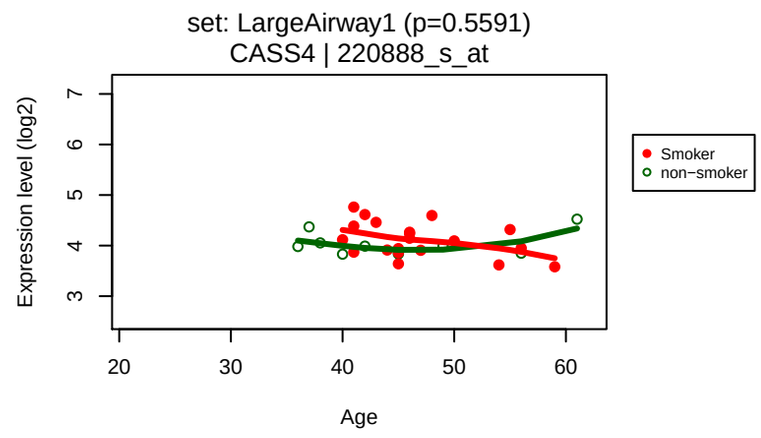

Gene expression of smokers and nonsmokers
gene: CASS4 | 220888_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1252
p=0.0891
p=0.0208
p=0.5591
p=0.5425
p=0.3152
p=0.4639
p=0.324
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

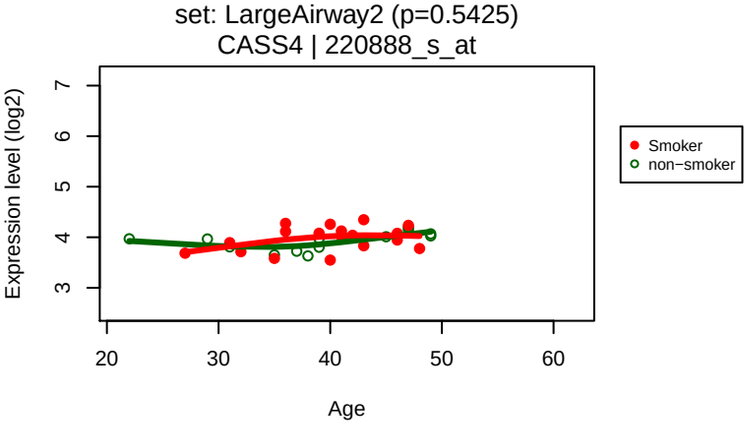

Gene expression ratio (smokers/nonsmokers)
gene: CASS4 | 220888_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1252
p=0.0891
p=0.0208
p=0.5591
p=0.5425
p=0.3152
p=0.4639
p=0.324
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

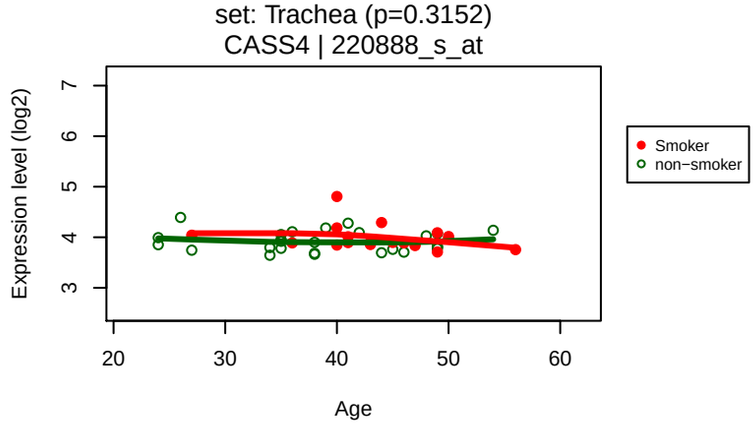

Gene expression ratio in Males
gene: CASS4 | 220888_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4244
p=0.0587
p=0.4056
p=0.8713
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CASS4 | 220888_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.228
p=0.0542
p=0.449
p=0.2323
SmallAirway2
LargeAirway2
Trachea
Macrophage2

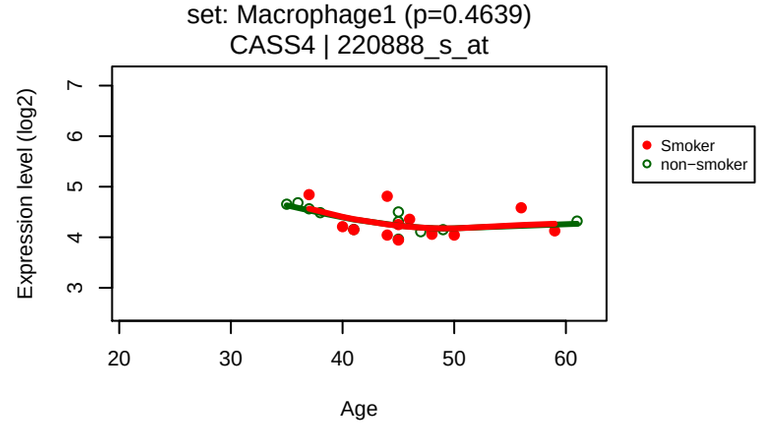

set: SmallAirway1 (p=0.1252)
CASS4 | 220888_s_at
Expression level (log2)
Age
Smoker
non−smoker

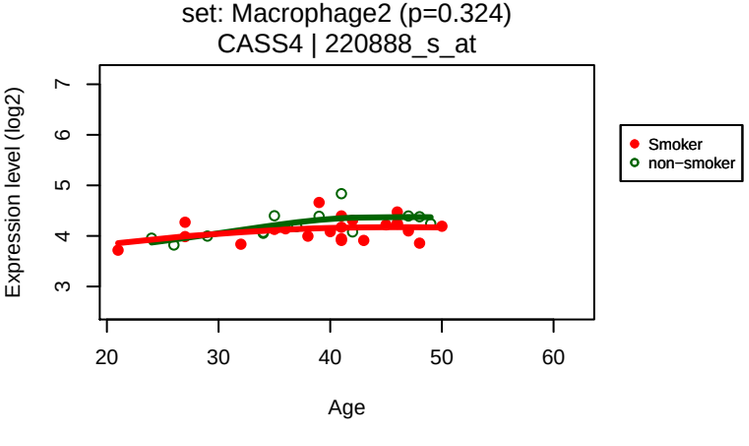

set: SmallAirway2 (p=0.0891)
CASS4 | 220888_s_at
Expression level (log2)
Age
Smoker
non−smoker

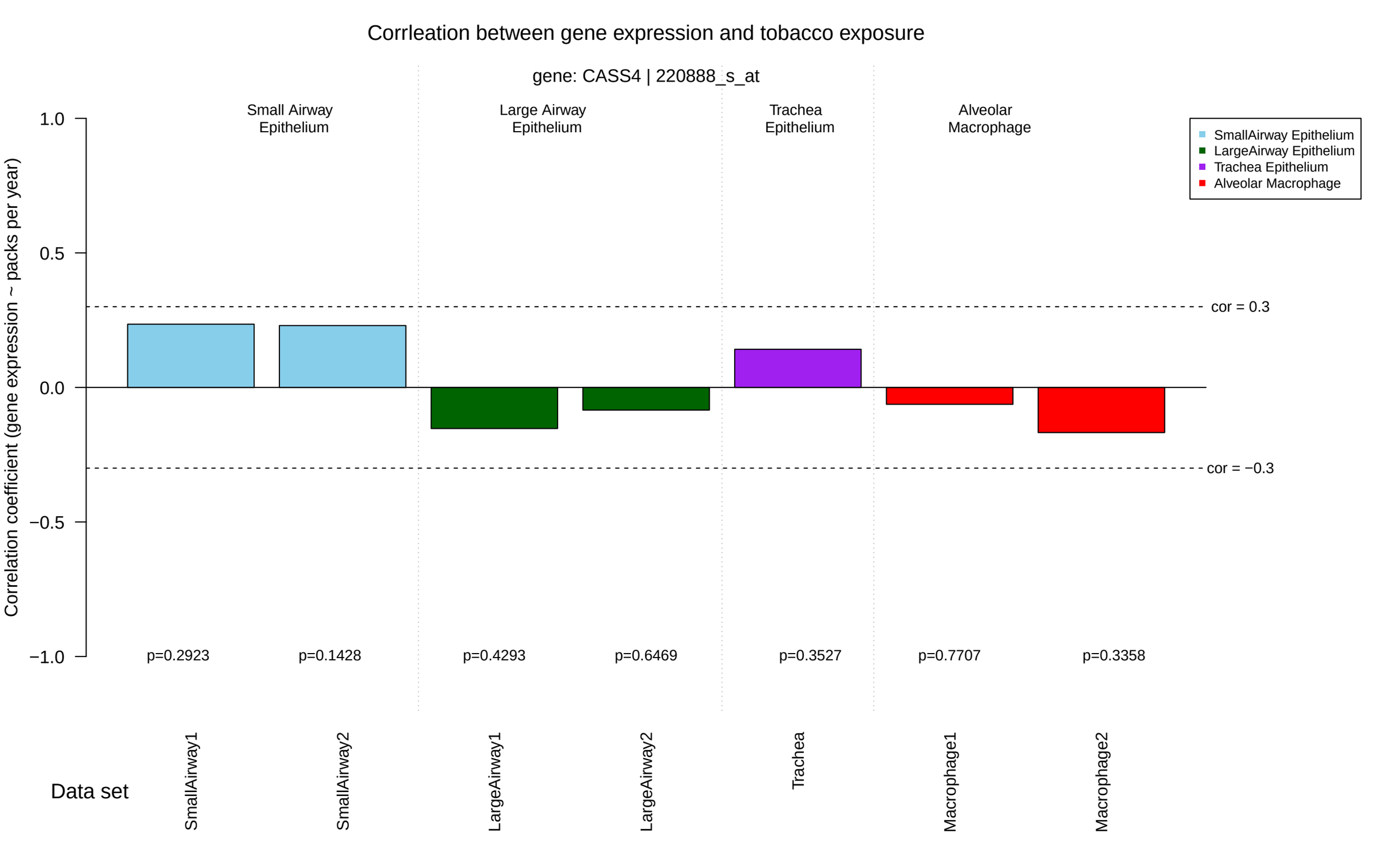

set: LargeAirway1 (p=0.5591)
CASS4 | 220888_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5425)
CASS4 | 220888_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3152)
CASS4 | 220888_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4639)
CASS4 | 220888_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.324)
CASS4 | 220888_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CASS4 | 220888_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2923
p=0.1428
p=0.4293
p=0.6469
p=0.3527
p=0.7707
p=0.3358
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2